

# Generating Useful Surface Defect Images When Real Data Is Scarce

Conditional GAN experiments on metal surface defects

SRUJAN VADDIPARTHI

SUPERVISOR: ABU ISLAM, SRI PRIYA DAS

GOLISANO INSTITUTE FOR SUSTAINABILITY

ROCHESTER INSTITUTE OF TECHNOLOGY

# Agenda

01

Why this problem  
matters

02

Research question

03

How a conditional  
GAN works

04

Experimental progression  
and results

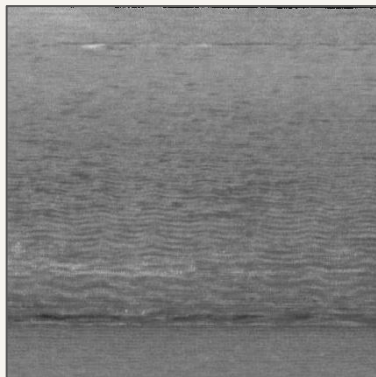
05

Key lessons, challenges, and  
next steps

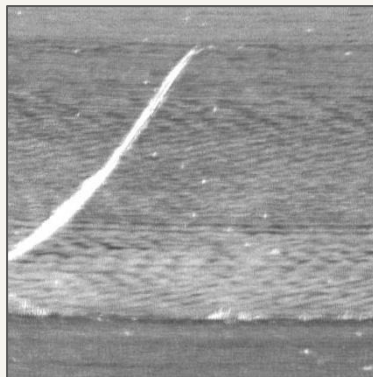
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# Why this problem matters

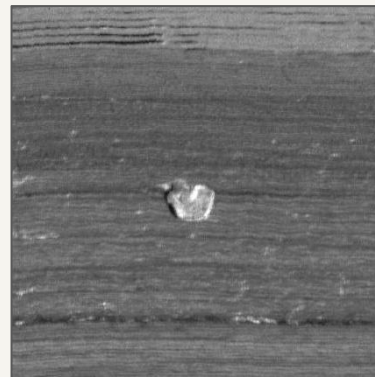
- Defect images are often scarce in industrial settings
  - Small datasets limit computer vision performance
  - Useful synthetic images could support inspection, reuse, and reduced waste
- 



Normal



Scratch



Spot

This is not a general image-generation problem. It is a low-data, defect-faithful industrial generation problem.

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# Research Question

Can AI generate useful defect images when real surface-defect data is scarce?

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What I wanted to understand:

Question 1

Can the model generate the defect type I ask for?

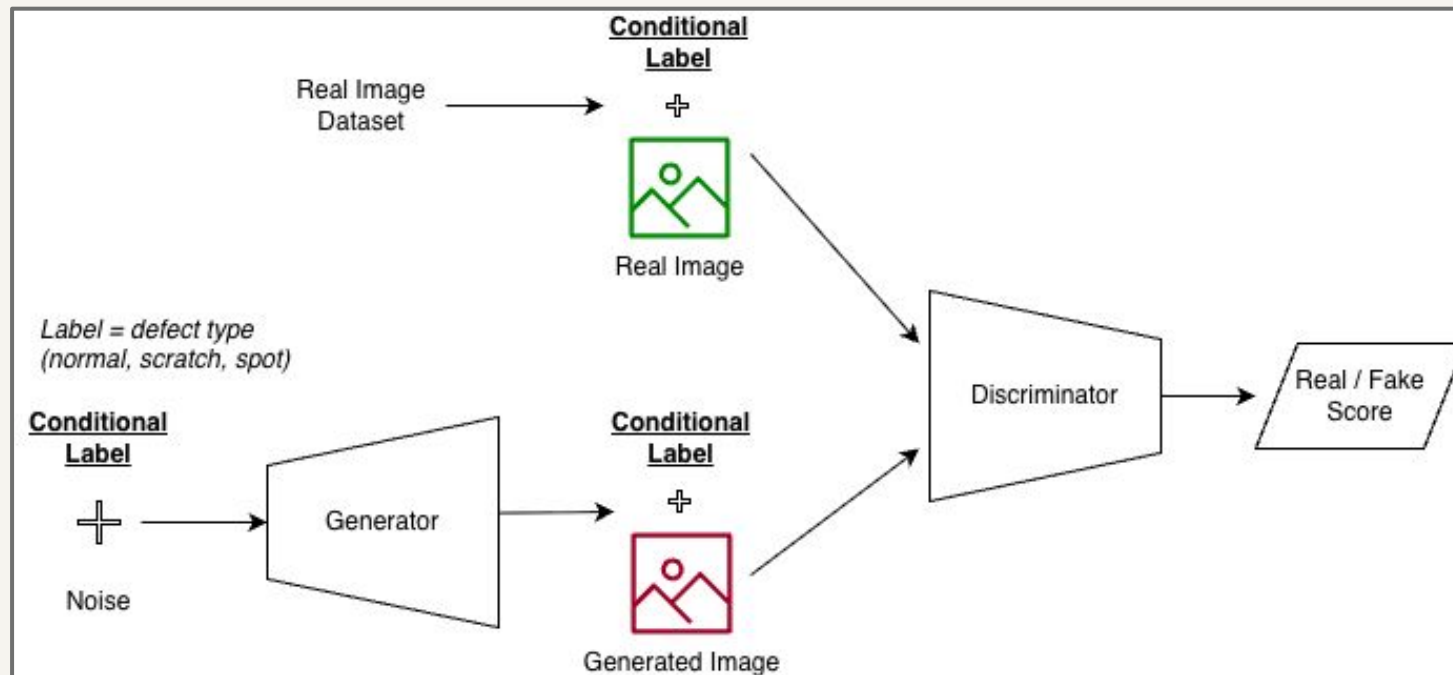
Question 2

Which defect types are easier to learn?

Question 3

What improves output quality when data is both scarce and imbalanced?

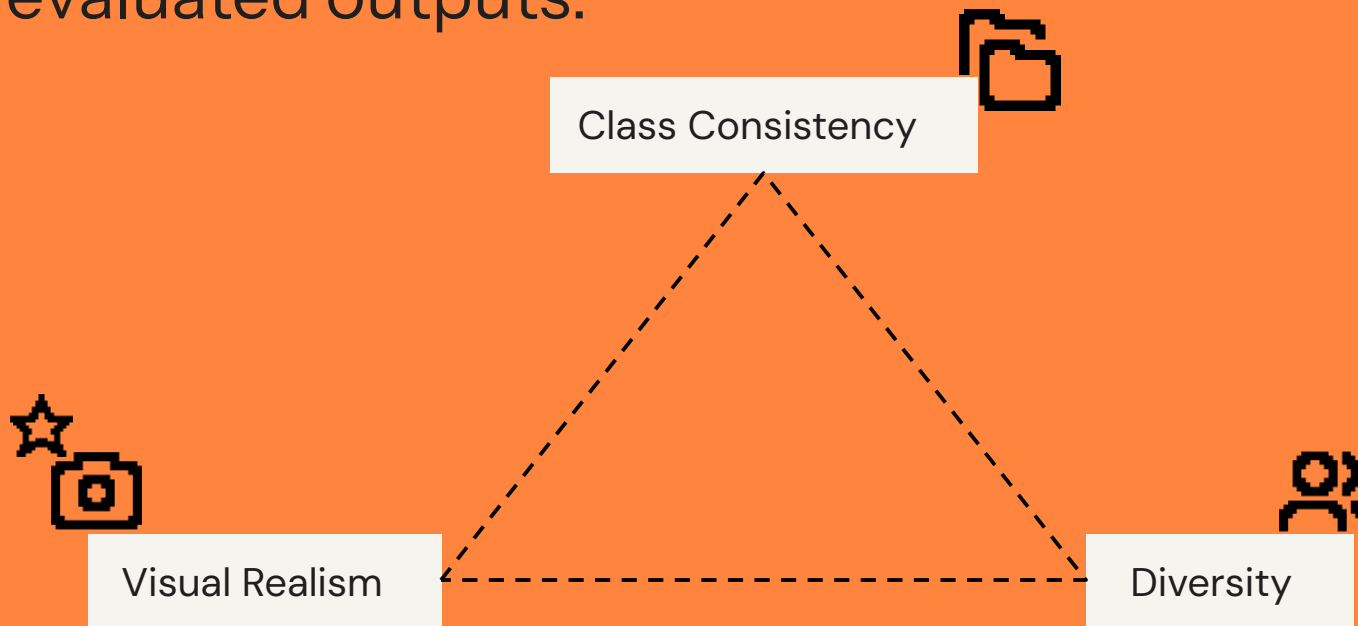
# How a Conditional GAN Works



The label steers what the generator creates and what the discriminator expects.

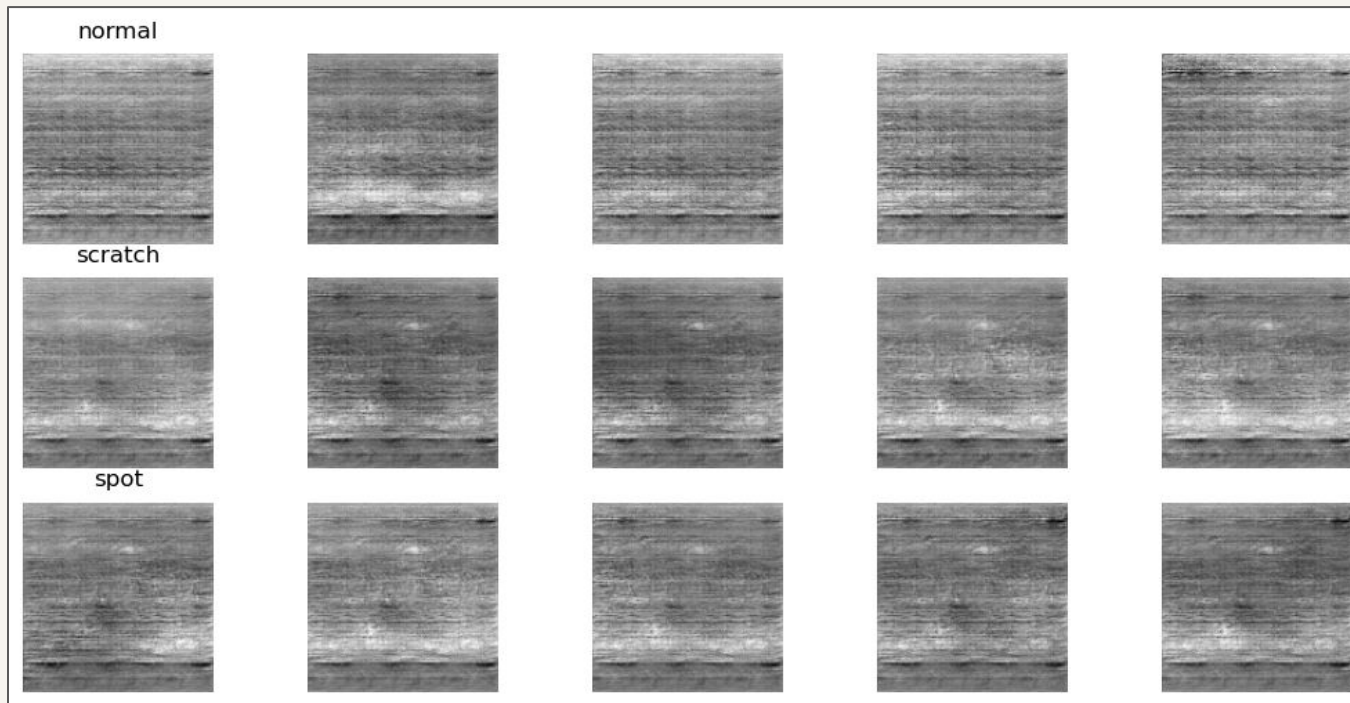
# How I Judged Progress

How I evaluated outputs:



# Starting Point: 3-Class Baseline

Generated Output:



- One cDCGAN for normal / scratch / spot
- Learned shared metal texture
- Weak defect-specific generation

*The 3-class setup  
learned metal  
texture better than  
defect structure.*

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# Why I Simplified the Problem?

From 3 classes to binary experiments.

-The 3-class setup weakened class-specific generation.

-I split the problem into:

- normal vs scratch
- normal vs spot

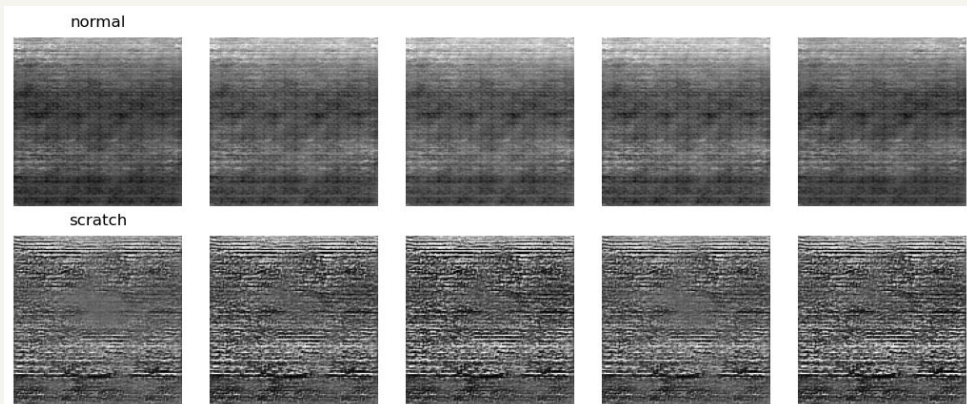
-I trained each setup for the same number of epochs, and also checked intermediate results.

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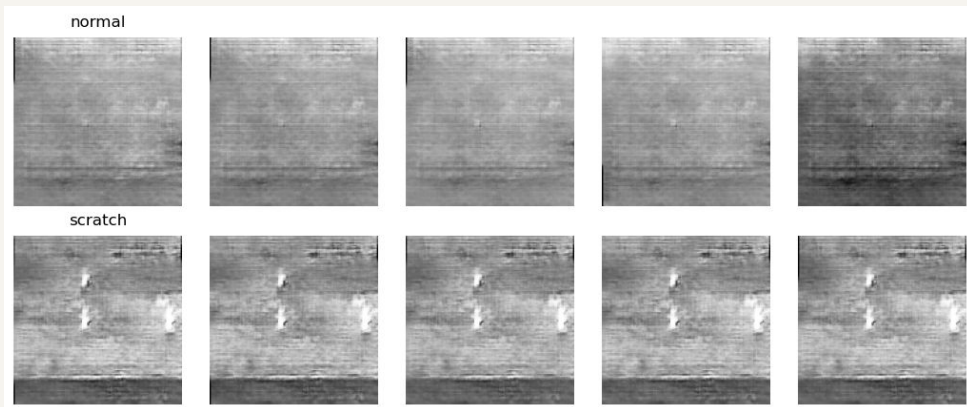


# Normal vs Scratch: What Changed

## Baseline Output:



## Final Setup Output

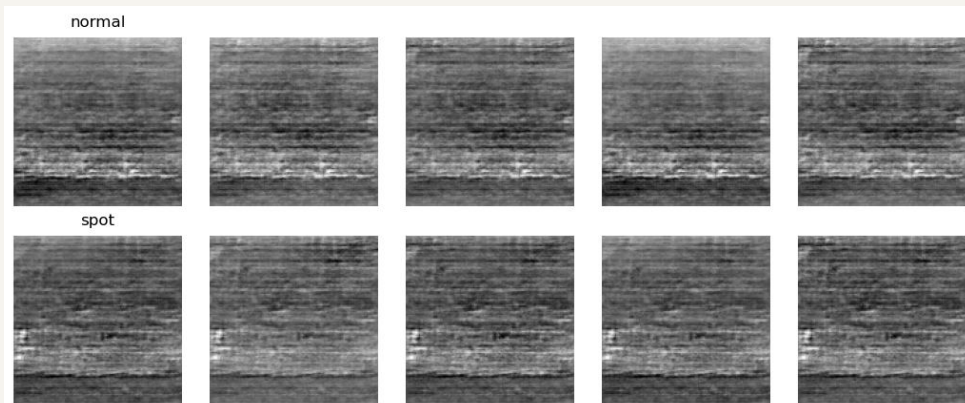


| Evaluation        | Judgement |
|-------------------|-----------|
| Visual Realism    | Medium    |
| Class Consistency | Medium    |
| Diversity         | Low       |

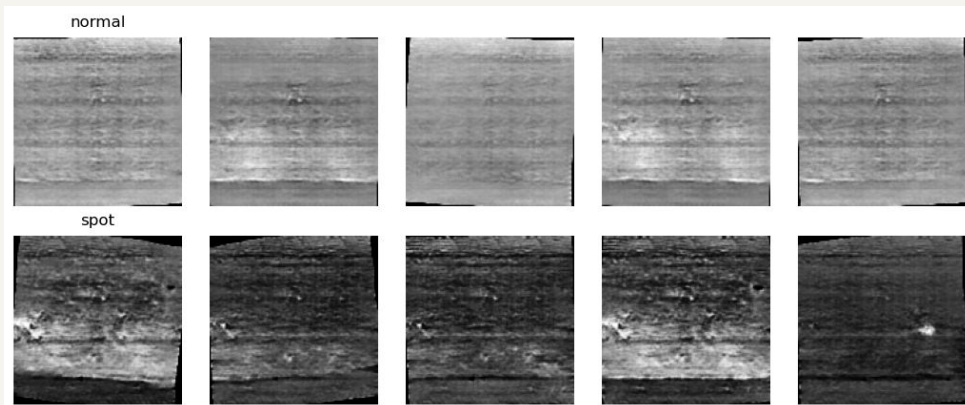
- Binary setup + enhancement improved scratch-vs-normal separation
- Diversity stayed low, and scratches were still hard to model

# Normal vs Spot: What Changed

## Baseline Output:



## Final Setup Output



| Evaluation        | Judgement |
|-------------------|-----------|
| Visual Realism    | Medium    |
| Class Consistency | High      |
| Diversity         | Low       |

- Spots were easier to learn than scratches
- Class consistency improved, but diversity stayed low

# What I learned

Repeated examples and limited surface variety strongly shaped what the model learned

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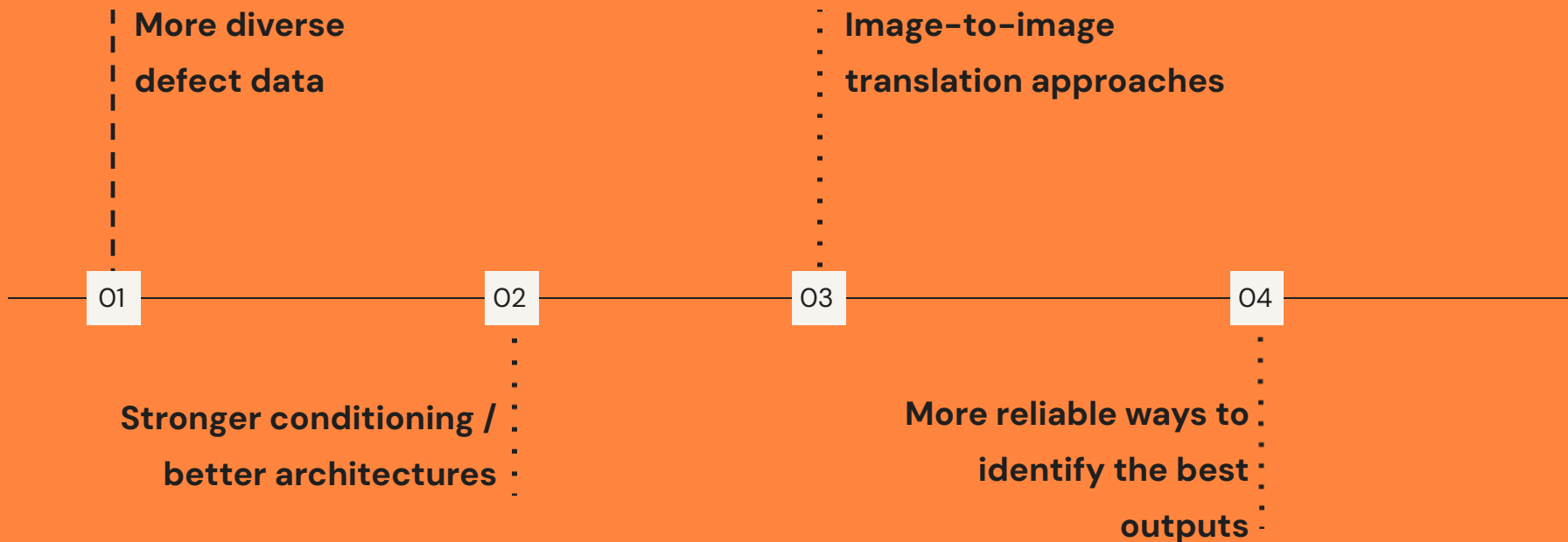
Spot-like defects were easier to generate than scratches

The model could separate classes better, but still repeated similar outputs

# Challenges

- 01 Structured defects were harder than localized ones
- 02 Narrow and repeated data reduced diversity
- 03 The current cDCGAN learned texture more easily than controlled defect structure
- 04 Best-looking outputs sometimes appeared before the final epoch

# What comes next



These findings suggest a path from low-data generation experiments toward more controllable and useful defect synthesis.

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# Conclusion

- The cDCGAN learned metal texture and some class-specific cues
- Spots were easier to learn than scratches
- Diversity remained the main limitation
- These results point clearly to the next set of experiments

# References / Acknowledgments

- Mirza & Osindero — Conditional Generative Adversarial Nets
- Radford et al. — DCGAN
- Isola et al. — pix2pix
- Zhu et al. — CycleGAN
- Defect-Transfer GAN paper
- Advisor: Abu Islam
- Co-mentor / support: Sri Priya Das
- RIT Research Computing

# Thank You ;)